

hovid Cefotaxime for Injection

DESCRIPTION

White to pale yellow crystalline powder.

COMPOSITION

hovid Cefotaxime for Injection 0.5 g:

Each vial contains:

Cefotaxime Sodium equivalent to Cefotaxime 0.5 g

hovid Cefotaxime for Injection 1 g:

Each vial contains:

Cefotaxime Sodium equivalent to Cefotaxime 1 g

hovid Cefotaxime for Injection 2 g:

Each vial contains:

Cefotaxime Sodium equivalent to Cefotaxime 2 g

ACTIONS AND PHARMACOLOGY

Cefotaxime is a third generation broad spectrum bactericidal cephalosporin antibiotic. The bactericidal properties are due to the inhibitory effect of cefotaxime on bacterial cell wall synthesis.

PHARMACOKINETICS

After a 1000mg intravenous bolus, mean peak plasma concentrations of cefotaxime usually range between 81 and 102 microgram/ml. Doses of 500mg and 2000mg produce plasma concentrations of 38 and 200 microgram/ml, respectively. There is no accumulation following administration of 1000mg intravenously or 500mg intramuscularly for 10 or 14 days.

The apparent volume of distribution at steady-state of cefotaxime is 21.6 litres/1.73m² after 1g intravenous 30 minute infusion.

Cerebrospinal fluid concentrations are low when the meninges are not inflamed, but are between 3 and 30 microgram/ml in children with meningitis. Cefotaxime usually passes the blood-brain barrier in levels above the minimum inhibitory concentration of common sensitive pathogens when the meninges are inflamed. Concentrations (0.2-5.4 microgram/ml), inhibitory for most Gram-negative bacteria, are attained in purulent sputum, bronchial secretions and pleural fluid after doses of 1 or 2g. Concentrations likely to be effective against most sensitive organisms are similarly attained in female reproductive organs, otitis media effusions, prostatic tissue, interstitial fluid, renal tissue, peritoneal fluid and gall bladder wall, after usual therapeutic doses. High concentrations of cefotaxime and desacetyl-cefotaxime are attained in bile.

Cefotaxime is partially metabolised prior to excretion. The principal metabolite is the microbiologically active product, desacetyl-cefotaxime. Most of a dose of cefotaxime is excreted in the urine - about 60% as unchanged drug and a further 24% as desacetyl-cefotaxime. Plasma clearance is reported to be between 260 and 390ml/minute and renal clearance 145 to 217 ml/minute.

After intravenous administration of cefotaxime to adults, the elimination half-life of the parent compound is 0.9 to 1.14 hours and that of the desacetyl metabolite, about 1.3 hours.

In neonates the pharmacokinetics are influenced by gestational and chronological age, the half-life being prolonged in premature and low birth weight neonates of the same age.

In severe renal dysfunction the elimination half-life of cefotaxime itself is increased minimally to about 2.5 hours, whereas that of desacetyl-cefotaxime is increased to about 10 hours. Total urinary recovery of cefotaxime and its principal metabolite decreases with reduction in renal function.

INDICATIONS

Treatment

Cefotaxime is indicated for the treatment of patients with serious infections caused by susceptible strains of the designated microorganisms in the diseases listed below.

- **Lower respiratory tract infections**, including pneumonia, caused by *Streptococcus pneumoniae* (formerly *Diplococcus pneumoniae*), *Streptococcus pyogenes** (Group A streptococci) and other streptococci (excluding enterococci, e.g., *Enterococcus faecalis*), *Staphylococcus aureus* (penicillinase and non-penicillinase producing), *Escherichia coli*, *Klebsiella* species, *Haemophilus influenzae* (including ampicillin resistant strains), *Haemophilus parainfluenzae*, *Proteus mirabilis*, *Serratia marcescens**, *Enterobacter* species, indole positive *Proteus* and *Pseudomonas* species (including *P. aeruginosa*).
 - **Genitourinary infections**. Urinary tract infections caused by *Enterococcus* species, *Staphylococcus epidermidis*, *Staphylococcus aureus**, (penicillinase and non-penicillinase producing), *Citrobacter* species, *Enterobacter* species, *Escherichia coli*, *Klebsiella* species, *Proteus mirabilis*, *Proteus vulgaris**, *Providencia stuartii*, *Morganella morganii**, *Providencia rettgeri**, *Serratia marcescens* and *Pseudomonas* species (including *P. aeruginosa*). Also, uncomplicated gonorrhea (cervicourethral and rectal) caused by *Neisseria gonorrhoeae*, including penicillinase producing strains.
 - **Gynecological infections**, including pelvic inflammatory disease, endometritis and pelvic cellulitis, caused by *Staphylococcus epidermidis*, *Streptococcus* species, *Enterococcus* species, *Enterobacter* species*, *Klebsiella* species*, *Escherichia coli*, *Proteus mirabilis*, *Bacteroides* species (including *Bacteroides fragilis**), *Clostridium* species, and anaerobic cocci (including *Peptostreptococcus* species and *Peptococcus* species) and *Fusobacterium* species (including *F. nucleatum**).
- Cefotaxime, like other cephalosporins, has no activity against *Chlamydia trachomatis*. Therefore, when cephalosporins are used in the treatment of patients with pelvic inflammatory disease and *C. trachomatis* is one of the suspected pathogens, appropriate anti-chlamydial coverage should be added.

- **Bacteremia/Septicemia** caused by *Escherichia coli*, *Klebsiella* species, and *Serratia marcescens*, *Staphylococcus aureus* and *Streptococcus* species (including *S. pneumoniae*).
- **Skin and skin structure infections** caused by *Staphylococcus aureus* (penicillinase and non-penicillinase producing), *Staphylococcus epidermidis*, *Streptococcus pyogenes* (Group A streptococci) and other streptococci, *Enterococcus* species, *Acinetobacter* species*, *Escherichia coli*, *Citrobacter* species (including *C. freundii**), *Enterobacter* species, *Klebsiella* species, *Proteus mirabilis*, *Proteus vulgaris**, *Morganella morganii*, *Providencia rettgeri**, *Pseudomonas* species, *Serratia marcescens*, *Bacteroides* species, and anaerobic cocci (including *Peptostreptococcus** species and *Peptococcus* species).
- **Intra-abdominal infections** including peritonitis caused by *Streptococcus* species*, *Escherichia coli*, *Klebsiella* species, *Bacteroides* species, and anaerobic cocci (including *Peptostreptococcus** species and *Peptococcus** species) *Proteus mirabilis**, and *Clostridium* species*.
- **Bone and/or joint infections** caused by *Staphylococcus aureus* (penicillinase and non-penicillinase producing strains), *Streptococcus* species (including *S. pyogenes**), *Pseudomonas* species (including *P. aeruginosa**), and *Proteus mirabilis**.
- **Central nervous system infections**, e.g., meningitis and ventriculitis, caused by *Neisseria meningitidis*, *Haemophilus influenzae*, *Streptococcus pneumoniae*, *Klebsiella pneumoniae** and *Escherichia coli**.

(* Efficacy for this organism, in this organ system, has been studied in fewer than 10 infections.

Although many strains of enterococci (e.g., *S. faecalis*) and *Pseudomonas* species are resistant to cefotaxime sodium in vitro, Cefotaxime has been used successfully in treating patients with infections caused by susceptible organisms.

Specimens for bacteriologic culture should be obtained prior to therapy in order to isolate and identify causative organisms and to determine their susceptibilities to Cefotaxime. Therapy may be instituted before results of susceptibility studies are known; however, once these results become available, the antibiotic treatment should be adjusted accordingly.

In certain cases of confirmed or suspected gram-positive or gram-negative sepsis or in patients with other serious infections in which the causative organism has not been identified, Cefotaxime may be used concomitantly with an aminoglycoside. The dosage recommended in the labeling of both antibiotics may be given and depends on the severity of the infection and the patient's condition. Renal function should be carefully monitored, especially if higher dosages of the aminoglycosides are to be administered or if therapy is prolonged, because of the potential nephrotoxicity and ototoxicity of aminoglycoside antibiotics.

It is possible that nephrotoxicity may be potentiated if Cefotaxime is used concomitantly with an aminoglycoside.

Prevention

The administration of Cefotaxime preoperatively reduces the incidence of certain infections in patients undergoing surgical procedures (e.g., abdominal or vaginal hysterectomy, gastrointestinal and genitourinary tract surgery) that may be classified as contaminated or potentially contaminated.

In patients undergoing cesarean section, intraoperative (after clamping the umbilical cord) and postoperative use of Cefotaxime may also reduce the incidence of certain postoperative infections. See DOSAGE AND ADMINISTRATION section.

Effective use for elective surgery depends on the time of administration. To achieve effective tissue levels, Cefotaxime should be given 1/2 or 1 1/2 hours before surgery. See DOSAGE AND ADMINISTRATION section.

For patients undergoing gastrointestinal surgery, preoperative bowel preparation by mechanical cleansing as well as with a non-absorbable antibiotic (e.g., neomycin) is recommended.

If there are signs of infection, specimens for culture should be obtained for identification of the causative organism so that appropriate therapy may be instituted.

To reduce the development of drug-resistant bacteria and maintain the effectiveness of Cefotaxime and other antibacterial drugs, Cefotaxime should be used only to treat or prevent infections that are proven or strongly suspected to be caused by susceptible bacteria. When culture and susceptibility information are available, they should be considered in selecting or modifying antibacterial therapy. In the absence of such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy.

CONTRAINDICATIONS

Hypersensitivity to cephalosporins.

In patients with a history of hypersensitivity to Cefotaxime and/or to any component of hovid Cefotaxime for Injection, a penicillin or to any other type of beta-lactam drug.

Allergic cross reactions can exist between penicillins and cephalosporins.

WARNINGS AND PRECAUTIONS

As with other antibiotics, the use of cefotaxime, especially if prolonged, may result in overgrowth of non susceptible organisms, such as *Enterococcus* spp., *Candida*, *Pseudomonas aeruginosa*. Repeated evaluation of the condition of the patient is essential. If superinfection occurs during treatment with cefotaxime, appropriate measures should be taken and specific anti-microbial therapy should be instituted if considered clinically necessary.

Cefotaxime has been associated with dizziness, which may affect the ability to drive or operate machinery.

Anaphylactic reactions: Preliminary enquiry about hypersensitivity to penicillin and other β -lactam antibiotics is necessary before prescribing cephalosporins since cross allergy occurs in 5-10% of cases. The use of cefotaxime is strictly contra-indicated in subjects with a previous history of immediate-type hypersensitivity to cephalosporins. Since cross allergy exists between penicillins and cephalosporins, use of the latter should be undertaken with extreme caution in penicillin sensitive subjects. Serious, including fatal hypersensitivity reactions have been reported in patients receiving cefotaxime. If a hypersensitivity reaction occurs, treatment must be stopped.

Serious bullous reactions: Cases of serious bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis have been reported with cefotaxime. Patients should be advised to contact their doctor immediately prior to continuing treatment if skin and/or mucosal reactions occur.

Patients with renal insufficiency: The dosage should be modified according to the creatinine clearance calculated. Patients with severe renal dysfunction should be placed on the dosage schedule recommended under "DOSAGE AND ADMINISTRATION".

Caution should be exercised if cefotaxime is administered together with aminoglycosides, probenecid or other nephrotoxic drugs. Renal function must be monitored in these patients, the elderly, and those with pre-existing renal impairment.

Haematological reactions: Leukopenia, neutropenia, and more rarely, agranulocytosis may develop during treatment with cefotaxime, particularly if given over long periods. For treatment courses lasting longer than 7-10 days, the blood white cell count should be monitored and treatment stopped in the event of neutropenia.

Some cases of eosinophilia and thrombocytopenia, rapidly reversible on stopping treatment, have been reported. Cases of haemolytic anaemia have also been reported.

Sodium Intake: The sodium content of cefotaxime should be taken into account when prescribing to patients requiring sodium restriction.

Clostridium difficile associated disease (e.g. pseudomembranous colitis): Cefotaxime may predispose patients to pseudomembranous colitis. If a diagnosis of pseudomembranous colitis is suspected, cefotaxime should be stopped immediately and appropriate specific antibody therapy should be started without delay.

Medicinal products that inhibit peristalsis should not be given.

Neurotoxicity: High doses of β -lactam antibiotics, including cefotaxime, particularly in patients with renal insufficiency, may result in encephalopathy (e.g. impairment of consciousness, abnormal movements and convulsions).

Patients should be advised to contact their doctor immediately prior to continuing treatment if such reactions occur.

Precautions for administration: Potentially life-threatening arrhythmia has been reported in a very few patients who received rapid intravenous administration of cefotaxime through a central venous catheter. The recommended time for injection or infusion should be followed.

PREGNANCY AND LACTATION

Pregnancy: The safety of cefotaxime has not been established in human pregnancy.

Cefotaxime crosses the placental barrier. Therefore, cefotaxime should not be used during pregnancy unless the anticipated benefit outweighs any potential risks.

Lactation: Cefotaxime passes into human breast milk in small amounts and is usually compatible with breast feeding, but careful monitoring of the infant is recommended.

Effects on the physiological intestinal flora of the breast-fed infant leading to diarrhoea, colonisation by yeast-like fungi, and sensitisation of the infant cannot be excluded.

Therefore, a decision must be made whether to discontinue breast-feeding or to discontinue therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

DRUG INTERACTIONS

Aminoglycoside antibiotics and diuretics: As with other cephalosporins, cefotaxime may potentiate the nephrotoxic effects of nephrotoxic drugs such as aminoglycosides or potent diuretics (e.g. furosemide). Renal function must be monitored.

Uricosurics: Probenecid interferes with renal tubular transfer of cefotaxime, thereby increasing cefotaxime exposure about 2-fold and reducing renal clearance to about half at therapeutic doses. Due to the large therapeutic index of cefotaxime, no dosage adjustment is needed in patients with normal renal function. Dosage adjustment may be needed in patients with renal impairment.

Interference with Laboratory Tests:

A false positive Coombs test may be seen during treatment with cephalosporins. This phenomenon may occur during treatment with cefotaxime and can interfere with blood cross-matching.

A false positive reaction to urinary glucose may occur with copper reduction methods (Benedict's, Fehling's or Clinistest) but not with the use of specific glucose oxidase methods.

There is a potential for mezlocillin and azlocillin to reduce the clearance of cefotaxime.

MAIN SIDE / ADVERSE EFFECTS

- Infections and Infestations**
Not known: Superinfection
- Blood and the lymphatic system disorders**
Uncommon: Leukopenia, Eosinophilia, Thrombocytopenia
Not known: Neutropenia, Granulocytopenia, Agranulocytosis, Haemolytic anaemia
- Immune system disorders**
Uncommon: Jarisch-Herxheimer reaction
Not known: Anaphylactic reactions, Angioedema, Bronchospasm, Anaphylactic shock
- Nervous system disorders**
Uncommon: Convulsions
Not known: Headache, Dizziness, Encephalopathy (e.g. impairment of consciousness, abnormal movements)
- Cardiac disorders**
Not known: Arrhythmia following rapid bolus infusion through central venous catheter
- Gastrointestinal disorders**
Uncommon: Diarrhoea
Not known: Nausea, Vomiting, Abdominal pain, Pseudomembranous colitis
- Hepato-biliary disorders**
Uncommon: Increase in liver enzymes (ALAT, ASAT, LDH, gamma-GT and/or alkaline phosphatase) and/or bilirubin
Not known: Hepatitis (sometimes with jaundice)
- Skin and subcutaneous disorders**
Uncommon: Rash, Pruritus, Urticaria, Drug fever
Not known: Erythema multiforme, Stevens-Johnson syndrome, Toxic epidermal necrolysis
- Renal and Urinary disorders**
Uncommon: Decrease in renal function/increase of creatinine (particularly when co-prescribed with aminoglycosides)
Not known: Interstitial nephritis, Candidiasis
- General disorders and administration site conditions**
Very common: For IM formulations: Pain at the injection site
Uncommon: Fever, Inflammatory reactions at the injection site, including phlebitis / thrombophlebitis
- Jarisch-Herxheimer reaction**
For the treatment of borreliosis, a Jarisch-Herxheimer reaction may develop during the first days of treatment.

The occurrence of one or more of the following symptoms has been reported after several week's treatment of borreliosis: skin rash, itching, fever, leucopenia, increase in liver enzymes, difficulty of breathing, joint discomfort.

OVERDOSE AND TREATMENT

Symptoms of overdose may largely correspond to the profile of side effects.

There is a risk of reversible encephalopathy in cases of administration of high doses of β -lactam antibiotics including cefotaxime.

In case of overdose, cefotaxime must be discontinued, and supportive treatment initiated, which includes measures to accelerate elimination, and symptomatic treatment of adverse reactions (e.g. convulsions).

No specific antidote exists. Serum levels of cefotaxime may be reduced by peritoneal dialysis or haemodialysis.

DOSAGE AND ADMINISTRATION

Adults

Dosage and route of administration should be determined by susceptibility of the causative organisms, severity of the infection, and the condition of the patient (see table for dosage guideline). Cefotaxime may be administered IM or IV after reconstitution. Premixed Cefotaxime Injection is intended for IV administration after thawing. The maximum daily dosage should not exceed 12 grams.

GUIDELINES FOR DOSAGE OF CEFOTAXIME

Type of Infection	Daily Dose (g)	Frequency and Route
Gonococcal urethritis/cervicitis in males and females	0.5	0.5 gram IM (single dose)
Rectal gonorrhea in female	0.5	0.5 gram IM (single dose)
Rectal gonorrhea in males	1	1 gram IM (single dose)
Uncomplicated infections	2	1 gram every 12 hours IM or IV
Moderate to severe infections	3-6	1-2 grams every 8 hours IM or IV
Infections commonly needing antibiotics in higher dosage (e.g., septicemia)	6-8	2 grams every 6-8 hours IV
Life-threatening infections	up to 12	2 grams every 4 hours IV

If *C. trachomatis* is a suspected pathogen, appropriate anti-chlamydial coverage should be added, because cefotaxime sodium has no activity against this organism.

To prevent postoperative infection in contaminated or potentially contaminated surgery, the recommended dose is a single 1 gram IM or IV administered 30 to 90 minutes prior to start of surgery.

Cesarean Section Patients

The first dose of 1 gram is administered intravenously as soon as the umbilical cord is clamped. The second and third doses should be given as 1 gram intravenously or intramuscularly at 6 and 12 hours after the first dose.

Neonates, Infants and Children

The following dosage schedule is recommended:

- Neonates (birth to 1 month):
 - 0-1 week of age: 50 mg/kg per dose every 12 hours IV
 - 1-4 weeks of age: 50 mg/kg per dose every 8 hours IV

It is not necessary to differentiate between premature and normal-gestational age infants.

- Infants and Children (1 month to 12 years):
 - For body weights less than 50 kg, the recommended daily dose is 50 to 180 mg/kg IM or IV body weight divided into four to six equal doses. The higher dosages should be used for more severe or serious infections, including meningitis. For body weights 50 kg or more, the usual adult dosage should be used; the maximum daily dosage should not exceed 12 grams.

Geriatric Use

This drug is known to be substantially excreted by the kidney, and the risk of toxic reactions to this drug may be greater in patients with impaired renal function. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function. (See PRECAUTIONS, General and PRECAUTIONS, Geriatric Use.)

Impaired Renal Function

NOTE: As with antibiotic therapy in general, administration of Cefotaxime should be continued for a minimum of 48 to 72 hours after the patient defervesces or after evidence of bacterial eradication has been obtained; a minimum of 10 days of treatment is recommended for infections caused by Group A beta-hemolytic streptococci in order to guard against the risk of rheumatic fever or glomerulonephritis; frequent bacteriologic and clinical appraisal is necessary during therapy of chronic urinary tract infection and may be required for several months after therapy has been completed; persistent infections may require treatment of several weeks and doses smaller than those indicated above should not be used.

Preparation of Cefotaxime Sterile

Cefotaxime for IM or IV administration should be reconstituted as follows:

Strength	Diluent (mL)	Withdrawable Volume	Approximate Concentration (mg/mL)
500 mg vial ^(*) (IM)	2	2.2	230
1 g vial ^(*) (IM)	3	3.4	300
2 g vial ^(*) (IM)	5	6.0	330
500 mg vial ^(*) (IV)	10	10.2	50
1 g vial ^(*) (IV)	10	10.4	95
2 g vial ^(*) (IV)	10	11.0	180
1 g infusion	50-100	50-100	20-10
2 g infusion	0-100	50-100	40-20

(*) in conventional vials

Shake to dissolve; inspect for particulate matter and discoloration prior to use. Solutions of CLAFORAN range from very pale yellow to light amber, depending on concentration, diluent used, and length and condition of storage.

For intramuscular use: Reconstitute VIALS with Sterile Water for Injection or Bacteriostatic Water for Injection as described above.

For intravenous use: Reconstitute VIALS with at least 10 mL of Sterile Water for Injection. Reconstitute INFUSION BOTTLES with 50 or 100 mL of 0.9% Sodium Chloride Injection or 5% Dextrose Injection. For other diluents, see COMPATIBILITY AND STABILITY section.

NOTE: Solutions of Cefotaxime must not be admixed with aminoglycoside solutions. If Cefotaxime and aminoglycosides are to be administered to the same patient, they must be administered separately and not as mixed injection.

A SOLUTION OF 1 G Cefotaxime IN 14 ML OF STERILE WATER FOR INJECTION IS ISOTONIC.

IM Administration: As with all IM preparations, Cefotaxime should be injected well within the body of a relatively large muscle such as the upper outer quadrant of the buttock (i.e., gluteus maximus); aspiration is necessary to avoid inadvertent injection into a blood vessel. Individual IM doses of 2 grams may be given if the dose is divided and is administered in different intramuscular sites.

IV Administration: The IV route is preferable for patients with bacteremia, bacterial septicemia, peritonitis, meningitis, or other severe or life-threatening infections, or for patients who may be poor risks because of lowered resistance resulting from such debilitating conditions as malnutrition, trauma, surgery, diabetes, heart failure, or malignancy, particularly if shock is present or impending.

For intermittent IV administration, a solution containing 1 gram or 2 grams in 10 mL of Sterile Water for Injection can be injected over a period of three to five minutes. Cefotaxime should not be administered over a period of less than three minutes. (See WARNINGS). With an infusion system, it may also be given over a longer period of time through the tubing system by which the patient may be receiving other IV solutions. However, during infusion of the solution containing Cefotaxime, it is advisable to discontinue temporarily the administration of other solutions at the same site.

For the administration of higher doses by continuous IV infusion, a solution of Cefotaxime may be added to IV bottles containing the solutions described below.

Compatibility and Stability

Solutions of Cefotaxime Sterile reconstituted as described above (Preparation of Cefotaxime Sterile) remain chemically stable (potency remains above 90%) as follows when stored in original containers and disposable plastic syringes:

Strength	Reconstituted Concentration mg/mL	Stability at or 22°C	Stability under Refrigeration at or below 25°C	Original Containers	Plastic Syringes
500 mg vial IM	230	12 hours	7 days	5 days	
1 g vial IM	300	12 hours	7 days	5 days	
2 g vial IM	330	12 hours	7 days	5 days	
500 mg vial IV	50	24 hours	7 days	5 days	
1 g vial IV	95	24 hours	7 days	5 days	
2 g vial IV	180	12 hours	7 days	5 days	
1 g infusion bottle	10-20	24 hours	10 days		
2 g infusion bottle	20-40	24 hours	10 days		

Reconstituted solutions stored in original containers and plastic syringes remain stable for 13 weeks frozen.

Reconstituted solutions may be further diluted up to 1000 mL with the following solutions and maintain satisfactory potency for 24 hours at or below 22°C, and at least 5 days under refrigeration (at or below 5°C): 0.9% Sodium Chloride Injection; 5 or 10% Dextrose Injection; 5% Dextrose and 0.9% Sodium Chloride Injection; 5% Dextrose and 0.45% Sodium Chloride Injection; 5% Dextrose and 0.2% Sodium Chloride Injection; Lactated Ringer's Solution; Sodium Lactate Injection (M/6); 10% Invert Sugar Injection, 8.5% Amino Acid Injection without Electrolytes.

Solutions of Cefotaxime Sterile reconstituted in 0.9% Sodium Chloride Injection or 5% Dextrose Injection in Vialflex® plastic containers maintain satisfactory potency for 24 hours at or below 22°C, 5 days under refrigeration (at or below 5°C) and 13 weeks frozen. DO NOT FREEZE.

NOTE: Cefotaxime solutions exhibit maximum stability in the pH 5-7 range. Solutions of Cefotaxime should not be prepared with diluents having a pH above 7.5, such as Sodium Bicarbonate Injection.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage: Store below 25°C. Protect from light and moisture.

Presentation/ Packing:
Vial of 0.5 g x 1's, 10's
Vial of 1 g x 1's, 10's
Vial of 2 g x 1's, 10's

Product Registration Holder/ Repacked by:

HOVID Bhd.
121, Jalan Tunku Abdul Rahman,
30010 Ipoh, Malaysia.
Manufactured by: NCPH Hebei Huamin Pharmaceutical Co., Ltd.
No. 98, Hainan Road, Economic & Technological Development Zone,
Shijiazhuang, China.

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